Task 1:

```
INCLUDE Irvine32.inc
.386
.model flat, stdcall
.stack 4096
ExitProcess proto, dwExitCode:dword

.data
	inputMsg	BYTE "Enter a number: ", 0
	sumMsg		BYTE "Sum: ", 0
	avgMsg		BYTE "Average: ", 0
	arr	DD 6 DUP(?)
	sum		DD 0
	avg	DD ?

.code

main PROC
	mov edi, OFFSET arr
	mov ecx, LENGTHOF arr

inputLoop:
	mov edx, OFFSET inputMsg
	call WriteString
	call ReadInt
	mov [edi], eax
	add edi, 4
loop inputLoop


	push LENGTHOF arr
	push OFFSET arr
	call getSum


	mov sum, eax


	push LENGTHOF arr
	push eax
	call getAvg


	mov avg, eax
```

```
        mov edx, OFFSET sumMsg
        call WriteString
        mov eax, sum
        call WriteInt
        call Crlf


        mov edx, OFFSET avgMsg
        call WriteString
        mov eax, avg
        call WriteInt
        call Crlf

        exit
main ENDP

getSum PROC
        push ebp
        mov ebp, esp

        mov esi, [ebp+8]
        mov ecx, [ebp+12]
        xor eax, eax

sumLoop:
        add eax, [esi]
        add esi, 4
loop sumLoop

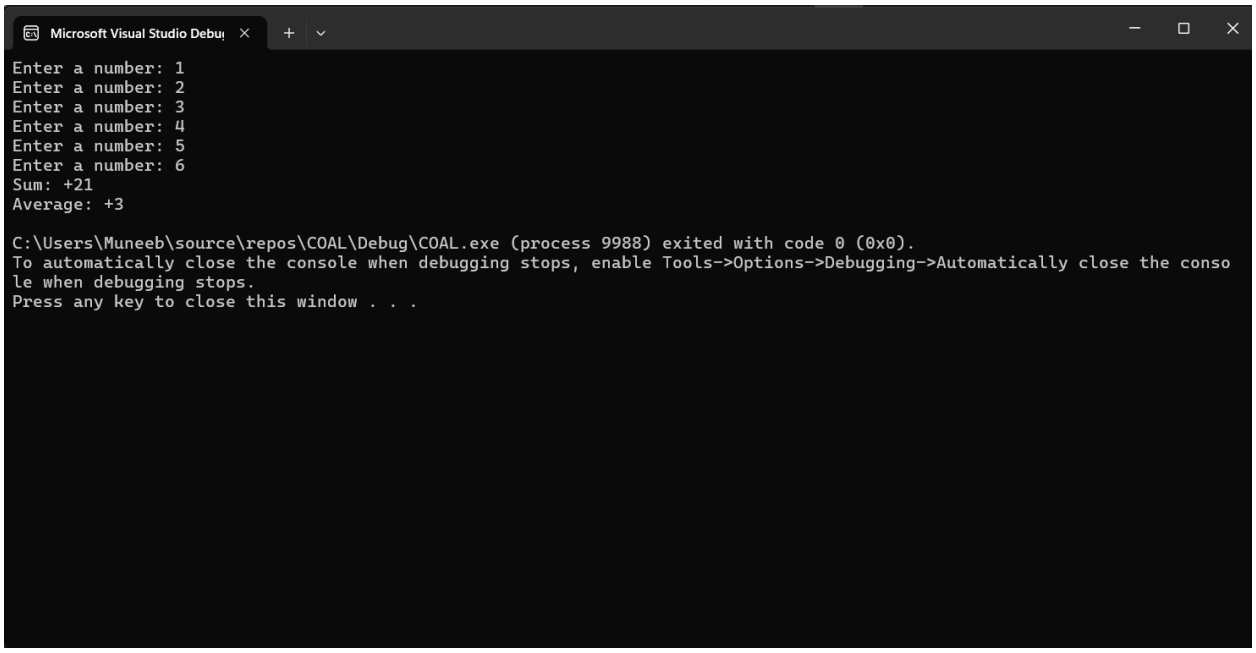
        pop ebp
        ret 8
getSum ENDP

getAvg PROC
        push ebp
        mov ebp, esp

        mov eax, [ebp+8]       ;sum
        xor edx, edx
        mov ebx, [ebp+12]   ;count
        div ebx
```

```
        pop ebp
        ret 8
getAvg ENDP

END main
```



Task 2:

```
include irvine32.inc
.386
.model flat,stdcall
.stack 4096
ExitProcess proto, dwExitCode:dword
.data

prompt1 BYTE "Enter a decimal number: ", 0
binaryArray BYTE 32 DUP(?)
value DWORD ?
index DWORD ?

.code

main proc

mov edx, OFFSET prompt1
call WriteString
```

```
call ReadInt
mov value, eax
mov ecx, 0
convertLoop:
mov eax, value
and eax, 1
add al, '0'
mov binaryArray[ecx], al
shr value, 1
inc ecx

cmp value, 0

jne convertLoop

dec ecx

printLoop:

mov al, binaryArray[ecx]

mov dl, al

call WriteChar

dec ecx

cmp ecx, -1

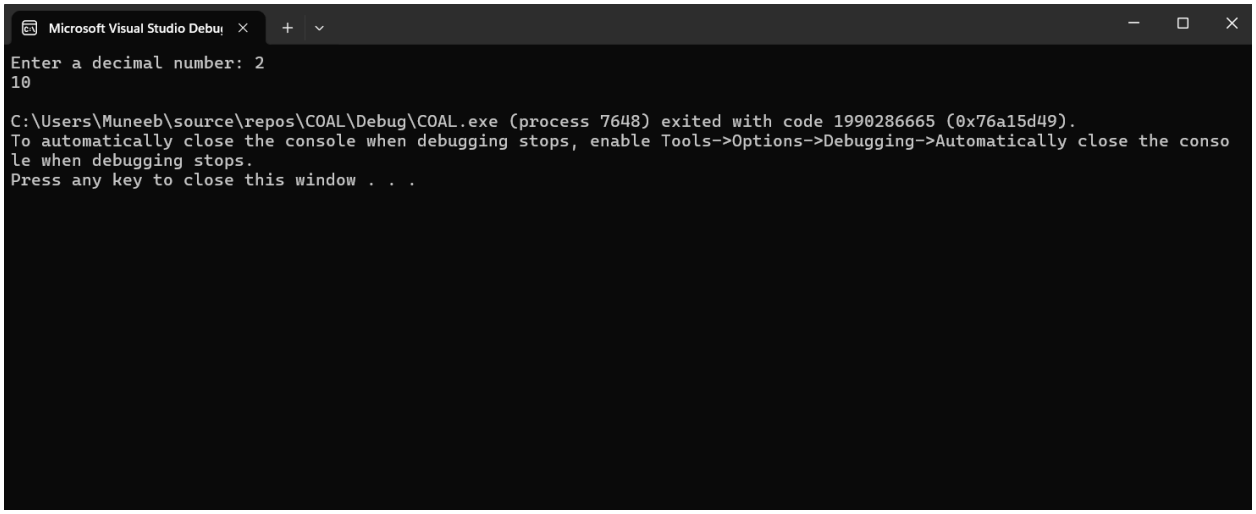
jne printLoop

call Crlf

call ExitProcess

main endp

end main
```



Task 3:

```
INCLUDE Irvine32.inc
.386
.model flat, stdcall
.stack 4096

.data
fileName        db "MyFile.txt", 0
fileHandle  dword ?
msgPrompt   db "Enter a number: ", 0
charInput   db ?
newline         db 13,10,0

.code
main PROC

        ; Create the file
        mov edx, OFFSET fileName
        call CreateOutputFile
        mov fileHandle, eax

        mov ecx, 5

InputLoop:
        ; Prompt user
        mov edx, OFFSET msgPrompt
        call WriteString


        call ReadChar
```

```
        mov charInput, al


        mov al, charInput
        call WriteChar
        call Crlf

        ; Write number to file
        mov eax, fileHandle
        mov edx, OFFSET charInput
        push ecx
        mov ecx, 1              ; write 1 byte
        call WriteToFile
        pop ecx

        loop InputLoop


        mov eax, fileHandle
        mov edx, OFFSET newline
        mov ecx, LENGTHOF newline - 1
        call WriteToFile

        ; Close file
        mov eax, fileHandle
        call CloseFile

        call ExitProcess
main ENDP

END main
```

Task 4:

```
INCLUDE Irvine32.inc
.386
.model flat, stdcall
.stack 4096

ExitProcess PROTO, dwExitCode:DWORD

.data
fileName        db "Grade.txt", 0
fileHandle  dword ?
msgPrompt   db "Expected grade in COAL lab: ", 0
```

```
grade           db "A", 0
buffer  db 2 dup(0)
newline         db 13, 10, 0

.code
main PROC

        mov edx, OFFSET fileName
        call CreateOutputFile
        mov fileHandle, eax

        mov edx, OFFSET msgPrompt
        call WriteString

        mov eax, fileHandle
        mov edx, OFFSET grade
        mov ecx, SIZEOF grade - 1
        call WriteToFile

        mov eax, fileHandle
        mov edx, OFFSET newline
        mov ecx, SIZEOF newline - 1
        call WriteToFile

        mov eax, fileHandle
        call CloseFile

        mov edx, OFFSET fileName
        call OpenInputFile
        mov fileHandle, eax


        mov eax, fileHandle
        mov edx, OFFSET buffer
        mov ecx, SIZEOF buffer
        call ReadFromFile


        mov edx, OFFSET buffer
        call WriteString
        call Crlf

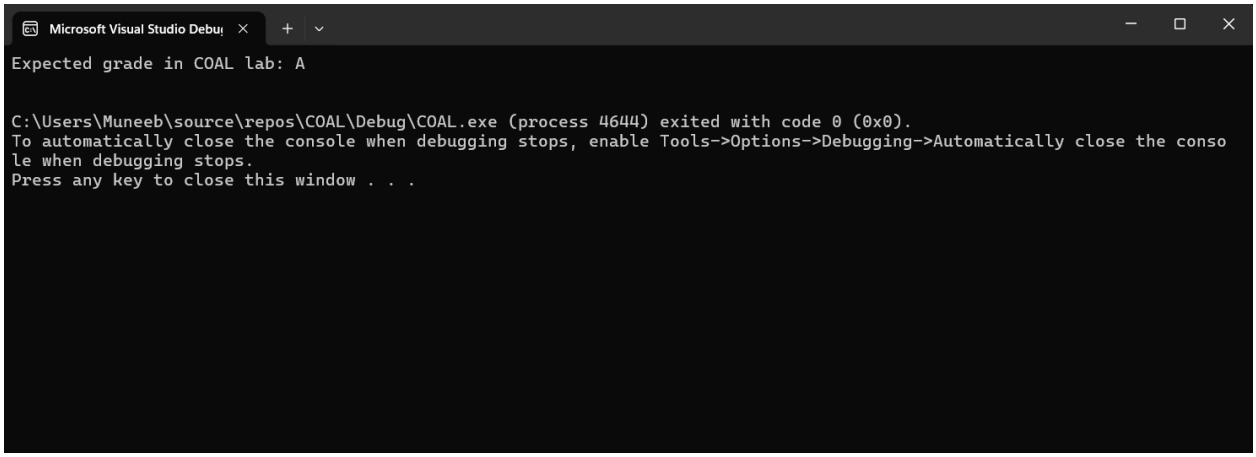
        mov eax, fileHandle
        call CloseFile
```

```
        jmp exit_program


exit_program:

        INVOKE ExitProcess, 0

main ENDP
END main
```



Task 5:

```
INCLUDE Irvine32.inc
.386
.model flat, stdcall
.stack 4096
ExitProcess proto, dwExitCode:dword

.data
        inputMsg        BYTE "Enter a number: ", 0
        sumMsg          BYTE "Sum: ", 0
        avgMsg          BYTE "Average: ", 0
        arr     DD 9 DUP(?)
        sum             DD 0
        avg     DD ?

.code

main PROC
        mov edi, OFFSET arr
        mov ecx, LENGTHOF arr
```

```
inputLoop:
	mov edx, OFFSET inputMsg
	call WriteString
	call ReadInt
	mov [edi], eax
	add edi, 4
loop inputLoop


	push LENGTHOF arr
	push OFFSET arr
	call getSum


	mov sum, eax


	push LENGTHOF arr
	push eax
	call getAvg


	mov avg, eax


	mov edx, OFFSET sumMsg
	call WriteString
	mov eax, sum
	call WriteInt
	call Crlf


	mov edx, OFFSET avgMsg
	call WriteString
	mov eax, avg
	call WriteInt
	call Crlf

	exit
main ENDP

getSum PROC
	push ebp
```

```
        mov ebp, esp

        mov esi, [ebp+8]
        mov ecx, [ebp+12]
        xor eax, eax

sumLoop:
        add eax, [esi]
        add esi, 4
loop sumLoop

        pop ebp
        ret 8
getSum ENDP

getAvg PROC
        push ebp
        mov ebp, esp

        mov eax, [ebp+8]       ;sum
        xor edx, edx
        mov ebx, [ebp+12]   ;count
        div ebx

        pop ebp
        ret 8
getAvg ENDP

END main
```

